# ZACKARY HORTON

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## Education

**Doctor of Philosophy**: September 2025-Present
**McGill University** – Montréal, QC
- Research focused degree studying algebraic number theory and automorphic forms
- Supervised by Professor Patrick Allen, Professor Henri Darmon

**Master of Science**: Pure Mathematics, September 2024-September 2025
**Imperial College London** - London, UK
- One year of full-time study along with a master's thesis project
- Classes in more advanced topics necessary to be able to expand deeper into research

**Bachelor of Science**: Honours Mathematics, Graduated May 2024
**McGill University** - Montréal, QC
- CGPA: 3.70/4.0
- Two-time Dean's list recipient
- Awarded honours with distinction upon graduating

## Research experience

**Master's thesis project:** Summer 2025
**Imperial College London**
- Studied local class field theory through a Galois cohomological lens
- Worked under the supervision of Professor Alexei Skorobogatov

**Undergraduate summer research:** Summer 2023
**McGill University**
- Awarded grant from the Natural Sciences and Engineering Research Council of Canada
- Studied modular forms and Serre's strong modularity conjecture
- Used SageMath to compute instances of non-minimal nebentypus
- Worked efficiently in a team of three people under the supervision of Professor Patrick Allen

**Undergraduate Project:** Winter 2024
**McGill University**
- Studied Quaternion Algebras by Voight to learn about tensors, algebras, maximal orders and connecting ideals
- Used this to produce a program that computed the graphs of maximal orders with connecting ideals as edges

## Extracurricular and other experience

**Teaching Assistant:** January 2026-Present
**McGill University**
- Practiced developing tutorials and problems to help with students' understanding
- Filled in for lectures when needed

**Math Tutor, McGill Math Help Desk:** 2023
**McGill University**
- Improved ability to communicate mathematical concepts
- Shared passion for math with others

**Airtanker Base Loader and Supervisor:** Summers 2020- 2022, 2024
**GNWT**
- Supervised employees and maintained air bases across the Northwest Territories
- Travelled across the territory to follow forest fire danger and mitigate risk of structure damage

**Team Member, Assistant Coach:** 2019-Present
**McGill Volleyball Team**
- Learned to thrive in competitive environment
- Developed team building and leadership skills

## Skills

- Language Skills: Native English, intermediate French
- Computer Skills: Proficiency with Python, SageMath, Java, C, and Unix-like systems for storage and computation